# GAN-ENHANCED DIFFUSION: BOOSTING SAMPLE QUALITY AND DIVERSITY

**Anonymous authors**
Paper under double-blind review

## ABSTRACT

Diffusion models have shown great promise in generating high-quality samples for various data types, but they often struggle with balancing sample fidelity and diversity. This trade-off is a common challenge in generative models due to their iterative nature. In this paper, we propose an enhanced diffusion model that integrates a Generative Adversarial Network (GAN) framework to address these challenges. We implement a simple discriminator network to distinguish between real and generated samples and modify the MLPDenoiser to include an adversarial loss term along with the existing reconstruction loss. Additionally, we introduce a gradient penalty to improve training stability. We validate our approach through extensive experiments on multiple 2D datasets, comparing the results in terms of training time, evaluation loss, KL divergence, and sample quality. Our results demonstrate that the GAN-enhanced diffusion model produces more realistic and diverse samples, achieving better performance across various metrics compared to baseline diffusion models.

CAUTION: THIS PAPER WAS AUTOMATICALLY GENERATED BY THE AI SCIENTIST

## 1 INTRODUCTION

Generative models have become a cornerstone of modern machine learning, with applications ranging from image synthesis to data augmentation. Among these, diffusion models have emerged as a powerful tool for generating high-quality samples across various data types (Ho et al., 2020). However, despite their success, diffusion models often face challenges related to sample quality and diversity.

The primary difficulty lies in balancing the trade-off between sample fidelity and diversity. High-fidelity samples may lack diversity, while diverse samples may suffer in quality. This trade-off is a common issue in generative models and is particularly pronounced in diffusion models due to their iterative nature (Yang et al., 2023).

In this paper, we propose an enhanced diffusion model that integrates a Generative Adversarial Network (GAN) framework to address these challenges. Our contributions are as follows:

- We implement a simple discriminator network to distinguish between real and generated samples, enhancing the sample quality.
- We modify the MLPDenoiser to include an adversarial loss term along with the existing reconstruction loss, improving the model's ability to generate realistic samples.
- We introduce a gradient penalty to the adversarial loss to improve training stability.
- We conduct extensive experiments on multiple 2D datasets to validate our approach, comparing the results in terms of training time, evaluation loss, KL divergence, and sample quality.

To verify our solution, we perform extensive experiments on multiple 2D datasets. We compare the results of our GAN-enhanced diffusion model with baseline diffusion models using various metrics, including training time, evaluation loss, KL divergence, and sample quality. Our results demonstrate that the GAN-enhanced diffusion model produces more realistic and diverse samples, achieving better performance across various metrics.

While our approach shows significant improvements, there are several avenues for future work. These include exploring more complex discriminator architectures, extending the model to higher-dimensional data, and investigating the impact of different adversarial loss functions.

## 2 Related Work

Generative models have seen significant advancements in recent years, with diffusion models and Generative Adversarial Networks (GANs) being two prominent approaches. In this section, we discuss the most relevant work in these areas and compare them with our proposed method.

Diffusion models, such as the Denoising Diffusion Probabilistic Model (DDPM) (Ho et al., 2020), have shown great promise in generating high-quality samples. These models work by reversing a diffusion process that gradually adds noise to the data. However, they often struggle with sample quality and diversity. The Elucidating the Design Space of Diffusion-Based Generative Models (EDM) (Karras et al., 2022) paper explores various design choices in diffusion models, providing insights into improving their performance. Our work builds on these insights by integrating a GAN framework to enhance sample quality.

GANs, introduced by Goodfellow et al. (2014), have been highly successful in generating realistic samples. They consist of a generator and a discriminator, where the generator aims to produce realistic samples, and the discriminator attempts to distinguish between real and generated samples. The integration of GANs with other generative models has been explored in various works (Tiago et al., 2024). For example, the work by Song et al. (2020) on Score-Based Generative Modeling through Stochastic Differential Equations demonstrates another approach to integrating GAN-like frameworks with diffusion models. For instance, the TabDDPM (Kotelnikov et al., 2022) paper combines diffusion models with GANs for tabular data, demonstrating the potential of this hybrid approach.

Our work differs in several key aspects. First, we focus on 2D datasets, which are more applicable to visual data, making our approach relevant for applications in image synthesis and related fields. Second, we introduce a gradient penalty to improve training stability, which is not commonly addressed in previous works. Third, we provide a comprehensive evaluation of our model's performance across multiple datasets, demonstrating significant improvements in sample quality and diversity. Unlike the TabDDPM (Kotelnikov et al., 2022) paper, which focuses on tabular data, our work is more applicable to visual data, making it relevant for applications in image synthesis and related fields.

In summary, while previous works have explored the integration of GANs with diffusion models, our approach is unique in its focus on 2D datasets, the introduction of a gradient penalty, and a comprehensive evaluation across multiple datasets. These contributions make our work a significant advancement in the field of generative models.

## 3 Background

Generative models have become a fundamental component of machine learning, enabling the creation of new data samples from learned distributions. These models have a wide range of applications, including image synthesis, data augmentation, and anomaly detection (Goodfellow et al., 2016).

Diffusion models are a class of generative models that generate data by reversing a diffusion process. This process involves gradually adding noise to the data and then learning to reverse this process to generate new samples. The Denoising Diffusion Probabilistic Model (DDPM) is a prominent example of this approach (Ho et al., 2020). Despite their success, diffusion models face challenges related to sample quality and diversity. The iterative nature of the diffusion process can lead to a trade-off between generating high-fidelity samples and maintaining diversity (Yang et al., 2023).

Generative Adversarial Networks (GANs) are another class of generative models that have shown remarkable success in generating high-quality samples. GANs consist of a generator and a discriminator, where the generator aims to produce realistic samples, and the discriminator attempts to distinguish between real and generated samples (Goodfellow et al., 2014). Integrating GANs with diffusion models can potentially address the challenges faced by diffusion models. By incorporating

AUTOMATICALLY GENERATED BY THE AI SCIENTIST

a discriminator network, the diffusion model can receive feedback on the realism of the generated samples, thereby improving sample quality.

### 3.1 Problem Setting

In this work, we aim to enhance the sample quality of diffusion models by integrating a GAN framework. Let $\mathbf{x}_0$ represent the original data, and $\mathbf{x}_t$ represent the data at timestep $t$ in the diffusion process. The goal is to learn a model that can generate $\mathbf{x}_0$ from $\mathbf{x}_t$ by reversing the diffusion process.

We assume that the diffusion process is defined by a noise schedule $\beta_t$, which controls the amount of noise added at each timestep. The model consists of a denoiser network $f_\theta$ and a discriminator network $D_\phi$. The denoiser network aims to reconstruct $\mathbf{x}_0$ from $\mathbf{x}_t$, while the discriminator network distinguishes between real and generated samples.

Our approach involves training the denoiser network with a combination of reconstruction loss and adversarial loss. The reconstruction loss ensures that the denoiser can accurately reverse the diffusion process, while the adversarial loss, provided by the discriminator, encourages the generation of realistic samples. We also introduce a gradient penalty to the adversarial loss to improve training stability.

## 4 Method

In this section, we present our approach to enhancing diffusion models by integrating a GAN framework. This method aims to improve sample quality by incorporating a discriminator network into the diffusion model training process. We detail the architecture of the denoiser and discriminator networks, the loss functions used, and the training procedure.

### 4.1 Denoiser Network

The denoiser network, denoted as $f_\theta$, reconstructs the original data $\mathbf{x}_0$ from the noisy data $\mathbf{x}_t$ at each timestep $t$. We employ a Multi-Layer Perceptron (MLP) architecture for the denoiser, which takes as input the noisy data and a sinusoidal embedding of the timestep. The network consists of an input layer, several residual blocks, and an output layer. The residual blocks capture complex patterns in the data, while the sinusoidal embeddings enable the network to effectively utilize temporal information.

### 4.2 Discriminator Network

The discriminator network, denoted as $D_\phi$, distinguishes between real and generated samples. We use a simple MLP architecture for the discriminator, which takes as input the data samples and outputs a probability score indicating the likelihood that the sample is real. The discriminator provides feedback to the denoiser, encouraging it to generate more realistic samples.

### 4.3 Loss Functions

Our training objective consists of two main components: the reconstruction loss and the adversarial loss. The reconstruction loss, $\mathcal{L}_{\text{recon}}$, ensures that the denoiser can accurately reverse the diffusion process. It is defined as the Mean Squared Error (MSE) between the predicted noise and the actual noise added to the data:

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_t, t}\left[\|f_\theta(\mathbf{x}_t, t) - \mathbf{n}\|^2\right], \tag{1}$$

where $\mathbf{n}$ is the noise added to the data.

The adversarial loss, $\mathcal{L}_{\text{adv}}$, encourages the denoiser to generate realistic samples. It is defined using the binary cross-entropy loss between the discriminator's predictions for real and generated samples:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{\mathbf{x}_0}\left[\log D_\phi(\mathbf{x}_0)\right] + \mathbb{E}_{\mathbf{x}_t}\left[\log(1 - D_\phi(f_\theta(\mathbf{x}_t, t)))\right]. \tag{2}$$

To improve training stability, we introduce a gradient penalty term, $\mathcal{L}_{\text{gp}}$, to the adversarial loss (Gulrajani et al., 2017). The gradient penalty is defined as:

$$\mathcal{L}_{\text{gp}} = \mathbb{E}_{\hat{\mathbf{x}}}\left[(\|\nabla_{\hat{\mathbf{x}}} D_\phi(\hat{\mathbf{x}})\|_2 - 1)^2\right], \tag{3}$$

CAUTION!!! THIS PAPER WAS AUTONOMOUSLY GENERATED BY THE AI SCIENTIST